

It is as if there is a certain operator (recall Dialogue Four) which generates $f'(x)$ at the output in response to $f(x)$ at the input. In other words, this operator, applied to the function $f(x)$, "generates" $f'(x)$. This operator is usually denoted by $\frac{d}{dx}$. This notation should be interpreted as a single entity and not as a ratio (it reads "d over dx").

Consider an "image" $\frac{d}{dx} \boxed{1} = \boxed{2}$. The squares in this expression symbolize the familiar "windows". "Window" 1 is to input $f(x)$, while "window" 2 outputs $f'(x)$. Thus, $\boxed{\frac{d}{dx} f(x) = f'(x)}$ (5)

Definition: The operation of obtaining $f'(x)$ from $f(x)$ is said to be the differentiation of $f(x)$.

The operator $\frac{d}{dx}$ performs this operation over $f(x)$ and is said to be the operator of differentiation.

Reader: But what exactly is $\frac{d}{dx}$ doing with $f(x)$?

Author: It is exactly the operation prescribed by (4). We may say that $\frac{d}{dx}$ "constructs" the ratio $\frac{f(x+\Delta x) - f(x)}{\Delta x}$ from $f(x)$ and determines the limit of this ratio (regarded as a function of Δx) at $\Delta x = 0$.

Reader: In other words, the operator $\frac{d}{dx}$ performs a certain limit transition operation, doesn't it?

Author: Certainly. The whole differential calculus (and with it, integral calculus) can be formulated in terms of certain limit transitions.

Reader: Why should we introduce an operator $\frac{d}{dx}$ if it represents nothing else but the limit transition operation described by (4)?

Author: You have posed a very important question. The problem is that if we had formulated differential calculus in terms of limits, using the relations of type (4), all books on calculus should have been increased in their volume several-fold and become hardly readable. The